SOCIAL HACKATHON

Theme: Student Innovation, Financial Inclusion & Social Good

We Win

AI-Driven Proactive Financial Orchestrator

System Architecture & Feature Documentation

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1 Problem Statement

1.1 Context: Why This Problem Matters Now

India's financial system has expanded access faster than it has expanded understanding. Bank accounts, UPI rails, and investment apps reach hundreds of millions of new users every year, but the guidance layer needed to use them well has not kept pace. The result is a country that is financially *included* but not financially *equipped*:

- **95%** of Indian individuals have no documented, structured financial plan covering emergency savings, insurance, debt, and retirement — most manage money reactively, month to month.
- Independent national surveys (NCFE) place financial literacy in India at roughly **27%**, well below the 55–75% levels seen in the US, Europe, and Australia.
- A single session with a traditional wealth advisor costs upwards of **Rs. 20,000**, placing structured, human financial planning permanently out of reach for the salaried middle class and gig economy alike.
- Roughly **3 out of every 4** retail investors admit to making emotion-driven decisions (panic-selling in downturns, herd-buying in rallies) rather than following a disciplined, goal-linked strategy.
- India's gig and platform workforce is projected to nearly **triple, from 7.7 million workers in 2020–21 to 23.5 million by 2029–30** (NITI Aayog). The overwhelming majority of this workforce has no CIBIL trail, and therefore no access to formal, low-cost credit — pushing them toward informal lenders and reinforcing cycles of debt.
- Existing banking and investment apps show a static account balance, not a forward-looking, obligation-aware number. Without predictive visibility into upcoming EMIs and bills, users routinely trigger overdraft penalties or miss payments — while idle surplus cash sits in low- or zero-yield accounts, quietly losing value to inflation.

1.2 The Banking & National Perspective

This is not only a consumer-side problem. Banks, NBFCs, and the wider financial system carry two structural inefficiencies of their own – and both ultimately trace back to the same absence of proactive, unified intelligence:

1. **Fragmented visibility into user activity across the app, web, and phone:** a customer's journey is scattered across the mobile app, the net-banking portal, IVR calls, and branch visits, with no single system stitching these touchpoints into one picture. A user who hits an error mid-loan-application on the app and immediately calls support is treated as a stranger by the call-centre agent, who has no visibility into what just happened. This fragmentation makes it nearly impossible for a bank to track, understand, or act on a user's real activity in its own portal – driving up Average Handling Time, repeated interrogation, and avoidable product drop-off at the exact moment a sale should close.
2. **No proper targeted advertising, and high middlemen costs specifically around loans:** banks and NBFCs spend heavily on generic, un-targeted advertising to attract loan and investment customers, and then pay Direct Selling Agents (DSAs) and other loan middlemen steep commissions to originate business – inflating Customer Acquisition Cost while still producing largely unqualified leads. This money is effectively spent in vain: it neither reaches the specific customers who would genuinely convert, nor does it help creditworthy but data-invisible applicants – especially gig and informal workers – get matched to the loans they actually qualify for.

The national and government dimension: these gaps compound into a macro-level problem. India's financial-inclusion push (Jan Dhan, UPI, Account Aggregator) has been extraordinarily successful at giving citizens *access* to the formal financial system, but access without guidance does not translate into genuine participation – trillions of rupees in household savings remain parked in physical assets (gold, real estate) or idle low-yield deposits rather than flowing into regulated capital markets, weakening the very capital-formation goals these policies were designed to support. At the same time, the credit-invisible gig and informal workforce is pushed toward unregulated informal lending, a channel that sits entirely outside RBI oversight and directly undermines national financial-stability and poverty-reduction objectives.

1.3 The Core Problem

Problem Statement

Indians lack access to intelligent, proactive financial guidance. Expert wealth advisors remain unaffordable to the vast majority of the population, digital banking and investment tools remain passive and generic dashboards rather than active advisors, and today's fintech solutions fail to understand an individual's evolving financial context or anticipate their changing needs. This gap leaves over 95% of the population managing money without a plan, exposes gig and informal workers to a permanent credit ceiling, and allows preventable losses — from overdraft penalties to inflation-eroded idle cash — to persist at national scale. On the institutional side, banks operate with a fragmented, cross-channel view of their own customers and rely on blind advertising and costly loan middlemen instead of precise, data-driven engagement – and at the national level, this combination leaves India's financial-inclusion agenda structurally incomplete: strong on access, weak on the guidance that would turn that access into real participation in the formal economy.

1.4 Objectives

Guided by this problem statement, **We Win** is built around eight concrete, measurable objectives:

1. **Democratize financial diagnosis:** deliver a zero-cost, deterministic, six-dimension Financial Health Score (Emergency Fund, Insurance, Investments, Debt, Tax, Retirement) to every user, replacing the Rs. 20,000-per-session advisor with an instant, RBI/SEBI/IT-Act-aligned assessment.

2. **Replace static balances with predictive cash intelligence:** surface a real-time *Safe-to-Spend* figure that accounts for upcoming EMIs, bills, and goals – not just the number currently in the account – to eliminate avoidable overdrafts and missed payments.

3. **Extend formal credit to the credit-invisible:** give India's fast-growing gig and informal workforce (projected to reach 23.5 million by 2029–30) a real-time, behaviour-based Alternative Credit Trust Score that proves repayment discipline without requiring a pre-existing CIBIL history.

4. **Turn idle cash into working capital:** automatically detect surplus balances and sweep them into regulated, higher-yield instruments, protecting household savings from inflation erosion at zero effort to the user.

5. **Guarantee explainable, hallucination-free guidance:** strictly separate deterministic financial computation (Python engines) from AI-generated explanation (LLM layer), so every number shown to a user is mathematically verifiable and every recommendation is under-

standable in plain English.

6. **Keep the plan alive, not static:** continuously ingest life events and market movements so the financial roadmap re-weights and re-optimizes itself in real time, rather than going stale the day it is generated.

7. **Create a virtuous B2B credit loop:** convert disciplined, verified user cashflow data into pre-qualified, low-risk loan leads for partner banks and NBFCs – cutting their customer acquisition cost while giving users faster, fairer access to credit.

8. **Unify the fragmented bank view and cut wasted loan spend:** stitch a customer's app, web-portal, and call-centre activity into a single real-time context graph, so partner banks can replace blind, generic advertising and expensive DSA/middlemen-driven loan origination with precise, pre-qualified engagement – reducing wasted marketing spend, Average Handling Time, and product drop-off in one motion.

2 Motivation

The motivation for We Win comes directly from structural failures we observed in India's personal finance ecosystem – on the user's side, the bank's side, and in how the two currently fail to communicate:

2.1 Expert Guidance Is a Privilege, Not a Right

With **95%** of Indians lacking a structured financial plan and traditional advisors charging upwards of **Rs. 20,000** per session, expert guidance remains reserved for the wealthy. The tools that could close this gap – AI, deterministic financial modelling, and cloud infrastructure – already exist; they have simply never been assembled into a single, trustworthy, zero-cost product for the mass market.

2.2 Digital Tools Are Passive, Not Proactive

Current digital finance apps provide static, generalized dashboards. Because they do not adapt to an individual's context, roughly **3 out of 4** investors end up making emotion-driven decisions, never building the financial literacy needed to take control of their own future. A system that actively monitors, explains, and nudges – rather than simply displaying numbers – can break this cycle.

2.3 Static Balances Create Preventable Financial Damage

Standard banking apps show a balance, not a forecast. Without active forecasting or auto-sweeps, users are left defenseless against emergencies, prone to accidental overdrafts, and watch idle cash quietly lose value to inflation while sitting in a low-yield savings account.

2.4 Credit Invisibility Traps the Informal Workforce

The absence of a CIBIL score denies gig and informal workers access to formal credit, pushing them toward high-interest informal lending. This does not just cost individuals money – it actively prevents them from ever building the credit history that would let them escape the trap, entrenching a cycle of poverty across a workforce projected to grow to 23.5 million people by 2030.

2.5 Users Leave Money They Already Earned on the Table

Most cardholders are sitting on unused, already-earned value – reward points about to expire, cashback never redeemed, lounge access never claimed, fee waivers never activated – simply because tracking these benefits across cards and accounts is inconvenient and no one proactively surfaces them. This is not a hypothetical gain; it is money and privileges the user has already qualified for. We Win watches each user's card and account benefits continuously and notifies them of unspent advantages before they lapse, converting invisible entitlements into real, realized value.

2.6 Relevant Bank and Business Offers Rarely Reach the Right Person

Banks and merchants do run genuinely valuable offers – pre-approved credit, fee waivers, partner cashback, upgrade paths – but blind, generic advertising means these offers rarely reach the specific user they would actually benefit. Because We Win already understands a user's real financial context, it can match relevant bank and business advantages to the right user as a timely, welcome suggestion rather than one more ignored promotional notification.

2.7 Change Should Be Explained, Not Just Applied

Life events and markets do not hold still, and a financial plan that does not move with them quickly becomes useless. When a new event or a market shock changes the numbers, We Win recalculates the user's Wealth Path in real time – but an unexplained change erodes user trust just as fast as a wrong one would. Every real-time adjustment is paired with a plain-English explanation from the XAI layer, so the user always understands *why* their path just changed and not just *that* it did, keeping them in control even as the system acts on their behalf.

Together, these observations convinced us that the right intervention is not another static calculator or a chatbot bolted onto a bank app, but a genuinely *proactive orchestrator* – one that watches, calculates, explains, and (with explicit user consent) acts, on behalf of both the user and the institutions serving them.

3 Proposed Solution: Features & Implementation

We Win is architected as a proactive orchestration layer that sits between the user and multiple financial institutions, powered by a strict separation between **deterministic financial computation** (never allowed to hallucinate) and an **AI explanation and orchestration layer** (Google Gemini, LangChain, CrewAI). Each feature below is described by what it does and how it is implemented.

3.1 Data Foundation: What We Store, and Where

Before any score or recommendation is generated, We Win needs a trustworthy, well-organized picture of the user's financial life. Rather than dumping every data point into one generic store, we split incoming data by *shape* across two purpose-built databases:

- **Supabase (PostgreSQL) – the high-velocity ledger.** This is where every raw, high-frequency, time-ordered signal lands: live bank cashflows pulled via MCP (salary credits, EMI debits, recurring bills, merchant categories), structured JSON extracted from uploaded documents (Form 16, Mutual Fund CAS statements), and every mobile app tap, web portal click, IVR call, and API event. Supabase stores this as an immutable, append-only stream – nothing is ever overwritten, only added, which is what makes the Safe-to-Spend calculation and the audit trail possible.
- **Neo4j – the identity & context graph.** Sitting directly on top of Supabase, Neo4j takes the fragmented, high-frequency logs and the user's qualitative inputs (risk tolerance, goals, target retirement age, life events) and maps them into a single relational knowledge graph per user – unifying their financial profile, goals, incidents, and cross-channel interactions into one traversable node.

This split is what lets every feature below stay fast on raw event lookups (Supabase) while still reasoning about *relationships* – how a life event affects a goal, or how a portfolio node maps to real holdings – through Neo4j.

3.2 1. Financial DNA Engine

What it does: Builds a complete, structured financial identity for the user by capturing cashflow, debts, assets, and goals, then scoring the user across six foundational dimensions:

- **Emergency Fund** – adequacy of accessible savings against monthly obligations
- **Insurance** – coverage adequacy across life, health, and asset protection
- **Investments** – diversification and alignment with goals and risk profile
- **Debt** – total leverage and quality of existing liabilities
- **Tax** – efficiency of current tax planning and eligible-benefit utilization
- **Retirement** – projected readiness against target retirement age and lifestyle

How we implemented it: Users upload key financial documents (salary slips, Form 16, Mutual Fund CAS statements); **PyMuPDF** extracts unstructured text and converts it into clean, structured JSON on our **FastAPI (Python)** backend, which is written into Supabase as described above. This JSON is fed into a fully **deterministic, hard-coded Python scoring engine** – never an LLM – that benchmarks each of the six dimensions against RBI, SEBI, and IT Act standards, guaranteeing an objective, nationally consistent Financial Health Score every time.

3.3 2. Autonomous Wealth Paths (Path Provisioning)

What it does: Once the Financial DNA Engine has scored the user, We Win generates three personalized wealth-building strategies – *Safe*, *Balanced*, and *Aggressive* – with one-click execution and automatic sweeping of idle cash into regulated, higher-yield instruments.

How we implemented it: Each Path is computed deterministically from three inputs: the user's Financial DNA score, their stated goals and risk profile, and a live, aggregated catalogue of investment and savings products pulled from *every* registered partner bank – not just one preferred partner. **This is the one service in the entire system where we deliberately surface products from all banks, because here the user's advantage matters more than any single business's advantage.** Once a user selects a Path, the Actionable Execution Engine (Section 4.5) drafts the required investment or sweep actions and routes them through mandatory Human-in-the-Loop approval before anything is executed. Progress against the chosen Path is tracked continuously and re-optimized whenever new cashflow, incident, or market data arrives.

3.4 3. Dynamic Incident Engine

What it does: Logs major life events – marriage, a new baby, a job change – and instantly re-weights financial priorities and the affected parts of the user's roadmap.

How we implemented it: Life events are written directly into the Neo4j knowledge graph, unifying the user's financial profile, goals, and event history into a single traversable node. This lets the engine instantly identify which parts of the roadmap are affected by a new event and re-trigger recalculation for just that portion – rather than re-running the entire plan from scratch.

3.5 4. Real-Time Market Intelligence

What it does: Filters market moves, bank policy changes, and SIP-related news down to only what is relevant to the user's actual holdings, and triggers portfolio recalibration when something material happens.

How we implemented it: Instead of ingesting the full market, our MCP layer first queries the user's Neo4j portfolio node to identify exactly which assets they hold, then runs scheduled, cost-effective checks against `yfinance` and `MFAPI` for only those holdings (see MCP Use #2, Section 4.4). A material move automatically feeds back into the Dynamic Incident Engine and the active Wealth Path for recalibration.

3.6 5. Cashflow Intelligence & the Safe-to-Spend Guardian

What it does: Continuously reads a user's live cashflow to protect and grow their money – and, in the same motion, gives partner banks the cashflow-based signal they need to serve that user better. We built this feature around two deliberately distinct advantages:

User Advantage:

- **Safe-to-Spend limits** – a real-time, obligation-aware balance (net of pending bills and EMIs) instead of a raw account number, so the user never overdraws by accident.
- **Unchecked autopays** – the engine flags recurring autopay debits the user has stopped actually using (a forgotten subscription, a lapsed service) before they silently drain money month after month.
- **Unused privileges of owning the product** – surfaces card and account benefits the user has already earned but not claimed (cashback, lounge access, fee waivers – see Section 2.5),

so ownership actually pays off.

- **Other user welfare** – idle-fund detection and auto-sweep into regulated, higher-yield instruments, protecting savings from inflation erosion at zero manual effort.

Bank / Sector Advantage:

- **Proper product suggestion** – because the engine already understands genuine cashflow behaviour, banks can recommend products that actually fit the user instead of relying on blind, generic advertising (Section 2.6).
- **Reducing costs** – proactively preventing overdrafts, missed EMIs, and support escalations lowers a bank's servicing and collections costs on that customer.
- **Ensuring welfare of the banks** – healthier, better-informed customers default less and stay longer, which directly improves portfolio quality and retention for the partner institution.

How we implemented it: Cashflow events streamed into Supabase are continuously netted – upcoming fixed obligations against incoming credits – to compute the live Safe-to-Spend threshold, with a proactive alert firing as the user approaches it. The same event stream is pattern-matched against the user's product holdings to detect stale autopays and unused card/account benefits, while aggregated, anonymized behavioural signals from this engine feed the product-suggestion and risk-reduction views banks receive.

3.7 6. Credit Trust Score & Contextual Loan Management

This feature combines two tightly linked ideas – first, a credit score built for people the formal system currently can't see, and second, the loan-matching engine that puts that score to work.

Part A – The Dual-Scoring Model. We deliberately compute *two* separate, deterministic scores rather than one – a general, cross-institution score and a bank-specific loyalty score – because "is this person creditworthy overall" and "how deep is this person's relationship with *this particular* bank" are genuinely different questions, and collapsing them into one number would either obscure global risk or unfairly punish a loyal customer at one bank for having no history at another.

A1. Standard Alternative Credit Trust Score (General).

- **What it is:** a real-time, behaviour-based credit score computed from cashflow discipline, income stability, and repayment history across *all* connected institutions – kept deliberately separate from the six-dimension Financial Health Score, because creditworthiness and overall financial wellness are related but not the same question.
- **Why it's needed:** a CIBIL score requires a pre-existing formal credit history, which excludes most of India's gig and informal workforce by definition. Without an alternative, financially disciplined earners with no loan history are treated identically to genuinely high-risk applicants, and are pushed toward informal, high-interest lenders instead.
- **How we calculate it:** entirely with deterministic math over *verified* signals – live cashflow regularity and repayment behaviour pulled via MCP (Section 4.3), never self-reported claims – and exposed as a structured, explainable score that partner banks can consume directly as a pre-qualified, low-risk lead.

A2. Bank Relationship Value (BRV) Score (Bank-Specific Loyalty).

- **What it is:** a separate score computed per partner bank, measuring the *depth of the user's engagement with that specific institution* – on-time repayment of loans held there, and idle or emergency funds parked in accounts there – rather than the user's global credit standing.

- **Why it's needed:** if a user consistently pays their SBI loan on time and keeps their emergency cash parked in SBI accounts, that loyalty is real value to SBI specifically, even if it says nothing about the user's behaviour at other institutions. A single global score cannot reward this without leaking bank-specific relationship data across the wider financial system.
- **How we calculate it:** deterministically, from the same verified MCP cashflow stream (Section 4.3), scoped per bank so that engagement with one institution never inflates or obscures the user's standing with another. A rising BRV Score at a given bank lets that bank waive processing fees or extend pre-approved, low-interest credit as a loyalty reward – without affecting the general Credit Trust Score any other bank sees.

Important: Scores Are a Supporting Signal, Not the Sole Basis for a Loan Decision

The Credit Trust Score and the BRV Score are deliberately built as *one strong, verifiable input* into a bank's underwriting decision – never as an automatic approval condition on their own. Cashflow discipline observed up to today is not a guarantee of behaviour after a loan is disbursed: a user's incentive to stay disciplined is naturally highest while they are still building credibility, and some borrowers relax that discipline precisely once a loan has been approved. For this reason, We Win positions both scores as a transparent, explainable signal that partner banks weigh alongside their own underwriting criteria, collateral checks, and risk policies – not as a replacement for a bank's independent judgment.

Part B – Contextual Loan Management.

- Behaves like a virtual loan officer: a rules-based detection layer watches the Supabase cashflow stream and the Neo4j-mapped incident log (Section 3.4) for patterns indicating a genuine credit need, such as a sudden expense spike or a life event.
- Loan options are then matched to the *specific* user – weighed against their Financial DNA score, their Credit Trust Score, and their real repayment capacity – rather than shown as generic offers.
- Users are not limited to reactive offers: they can also ask the system custom, hypothetical questions ("What if I took a Rs. 3 lakh loan for a bike right now?") and get the impact on their Safe-to-Spend limit, debt ratio, and Wealth Path explained back in simple, plain-English terms through the XAI layer (Section 3.8) – letting them explore options safely before ever applying.
- An optimization routine mathematically re-routes existing debt toward the fastest path to zero balance, rather than the bank-favoured minimum-payment default.

3.8 7. Explainability (XAI) Layer

What it does: Translates every complex financial calculation and recommendation into a simple, accurate, plain-English explanation the user can trust.

How we implemented it: All financial *logic* runs in deterministic Python – the AI layer (Google Gemini, orchestrated via LangChain and CrewAI) is only ever shown the already-computed, verified numbers and is prompted strictly to explain *why* a metric changed or a strategy was recommended. Because the AI never performs the arithmetic itself, it cannot hallucinate a financial figure – it can only narrate one that has already been verified.

4 MCP and Agents

4.1 Multi-Agent Workflow

We Win's AI layer is not a single monolithic LLM call – it is a coordinated crew of narrowly-scoped agents, orchestrated with **CrewAI** and bound to backend and external tools through **LangChain**, with **Google Gemini** as the underlying model for every agent. Each agent owns exactly one responsibility and is only ever handed the specific MCP tools it needs to do that job:

- **Aggregator Agent** – calls the Multi-Bank Aggregation MCP tool (MCP Use #1) on a schedule and on-demand, pulling live cashflow, EMI, and bill data into the system for the Financial DNA Engine and the Safe-to-Spend Guardian.
- **Context & Incident Agent** – watches for newly logged life events and cross-channel journey signals, pushes structured updates into the Neo4j graph, and flags which parts of the roadmap need recalculation (Section 3.4).
- **Market Watch Agent** – calls the Selective Market Connectivity MCP tool (MCP Use #2), fetching prices only for the user's actual holdings, and raises a recalibration event when a material move is detected.
- **Loan Matching Agent** – calls the Automated Bank Policy Parsing MCP tool (MCP Use #4) whenever the Contextual Loan Management engine detects a genuine credit need, and ranks the resulting eligible offers against the user's Credit Trust and BRV Scores.
- **Explainer (XAI) Agent** – fetches the correct, pre-approved explanation template via MCP Prompts (MCP Use #5) before generating any plain-English narration of a score, alert, or recommendation.
- **Execution Agent** – assembles the Drafted Action Payload once a user selects a Wealth Path action or loan option, and calls the Actionable Execution MCP tool (MCP Use #3) to record it and, on approval, route it onward – see the bidirectional-write pattern described under MCP Use #3.

Agents never touch the database. Regardless of role, no agent is ever granted a direct database connection, credential, or query interface to Supabase or Neo4j. Because the data involved is financial, the bar for integrity and confidentiality is high – a hallucinated or malformed query issued directly by an LLM-driven agent could corrupt a ledger or leak one user's records into another's context. Every read and every write an agent performs is instead mediated through a narrowly-scoped MCP tool call, so the agent only ever sees the exact, pre-validated slice of data a tool is designed to return, and can only write through a tool designed to accept exactly that write – never raw table access in either direction.

4.2 Effective Use of the Model Context Protocol (MCP)

The Model Context Protocol is the backbone that lets these agents act as a genuine *orchestrator* across banks, markets, and execution systems, rather than a siloed app calling raw APIs. We deliberately use MCP in five high-leverage places, each chosen because it needs live, structured, tool-callable access to an external system – or a controlled, auditable way to reach the database – rather than a one-time data dump or direct access.

4.3 MCP Use #1 — Multi-Bank Data Aggregation

Instead of asking users to manually upload statements for every account, MCP connects to a user's accounts across single or multiple registered banks and pulls live financial state directly – salary credits, EMI debits, recurring bills, and merchant categories. This live stream feeds

both the Financial DNA Engine and the Safe-to-Spend Guardian, replacing stale, self-reported data with verified, real-time ground truth. **Relevance:** this is also the primary place where the "agents never touch the database" principle earns its keep – the Aggregator Agent never holds bank credentials or a direct account-data query interface; it only ever receives the pre-validated, scoped payload the MCP tool chooses to return, which is what lets us guarantee financial-grade integrity and confidentiality even as more banks are added.

4.4 MCP Use #2 — Selective, Context-Aware Market Connectivity

Rather than ingesting the entire stock and mutual fund market, our MCP layer first queries the user's Neo4j portfolio node to identify *exactly* which holdings they own, then calls `yfinance` and `MFAPI` *only* for those assets. This keeps the Real-Time Market Intelligence feature (Section 3.5) cost-effective (scheduled daily checks instead of a live feed) while ensuring every market shock or price movement relevant to the user is caught and used to trigger portfolio recalibration.

4.5 MCP Use #3 — Actionable Execution with Human-in-the-Loop Guardrails

MCP tool definitions expose the operational capabilities needed to actually act on a recommendation – generating a stock/mutual fund buy or sell order, or preparing a loan application payload. Critically, these tools are never invoked autonomously: the Execution Agent only ever produces a **Drafted Action Payload**, which is shown to the user in plain English and executed *only* after explicit 1-click biometric or OTP approval. **Relevance – bidirectional MCP flow:** we do not use MCP purely to *read* data. Every drafted and approved action is also *written back* through MCP into a dummy, immutable ledger – modelled on a real bank transaction ledger – so the full lifecycle of an action (drafted, approved, executed) is captured as an append-only record before it ever touches a live bank system. This bidirectional read/write pattern is what makes MCP the genuine execution arm of the system, rather than a one-way data feed, while keeping the human unambiguously in control of every binding financial action.

4.6 MCP Use #4 — Automated Bank Policy Parsing for Contextual Loan Matching

When the Contextual Loan Management feature detects a genuine credit need, MCP is used to fetch and parse the complex loan documents and policy terms of partner banks in real time, extracting the interest rates and eligibility terms relevant to the user's specific profile. This eliminates manual middleman friction and lets the system surface pre-approved, genuinely comparable loan options the moment a need is detected – rather than generic marketing offers.

4.7 MCP Use #5 — Standardized Interactions via MCP Prompts

MCP servers do not just expose data and execution tools – they can also host predefined **prompt templates**. This guarantees that whenever an agent interacts with a specific domain, it uses the optimal, most secure instructions defined by the server, rather than whatever prompt happens to be hardcoded on the client. **How we use it:** our Explainable AI (XAI) instructions are stored as an MCP Prompt on the backend. When the Next.js frontend needs to explain a financial score or a recommended action, the Explainer Agent requests this specific prompt template from the MCP server instead of building it locally. This keeps our system prompts centralized, version-controlled, and modular – a prompt can be audited, updated, or rolled back in one place – rather than duplicated and hardcoded into client-side application code.

5 Security Procedures & Strategies

Because We Win handles sensitive financial data and can draft real monetary actions, security is treated as a first-class architectural constraint, not an afterthought. Our approach rests on five pillars:

5.1 Deterministic-Core, AI-Explains-Only Design

All financial arithmetic – scores, safe-to-spend limits, debt optimization, credit scores – runs in hard-coded, testable Python engines. The AI layer (Gemini via LangChain/CrewAI) is architecturally restricted to explanation and orchestration and is never permitted to perform or alter a financial calculation. This eliminates AI hallucination as a source of financial risk by construction, not by prompting.

5.2 Mandatory Human-in-the-Loop (HITL) Approval

No binding financial action – an investment order, a loan application, a cash sweep – is ever executed autonomously. Every action is first materialized as a Drafted Action Payload and pushed to the user in plain English; execution requires explicit 1-click biometric or OTP confirmation. This guarantees the user is always the final authority on their own money.

5.3 Least-Privilege, Purpose-Scoped MCP Access

Every MCP connection is scoped to the minimum data a feature actually needs – multi-bank aggregation reads cashflow, never triggers payments; market MCP calls are restricted to the user's real holdings, never the full market; and the Autonomous Wealth Paths feature is the *only* service permitted to pull the full, all-bank product catalogue, precisely because that is the one place a wider view directly serves the user rather than any single partner's interest. Scoping access this tightly at the tool-definition level, rather than trusting the AI layer to self-restrict, keeps a compromised or misbehaving component from reaching data or actions outside its designated purpose.

5.4 Data Minimization & Internal Processing

All sensitive computation – scoring, credit assessment, cashflow forecasting – is processed internally within our own backend. No user financial data is shared with or exposed to external third parties beyond the minimum, purpose-scoped calls required to fetch live bank or market data via MCP, and every such call is scoped to only the data required for that specific feature (e.g., market MCP calls are restricted to the user's actual holdings, never a full market dump).

5.5 Immutable Audit Trail & Compliance Alignment

Supabase's append-only, time-series ledger design means every event – a tap, a transaction, an API timeout, a drafted action – is permanently recorded, giving us a complete, tamper-resistant audit trail for every recommendation and every executed action. All scoring logic is benchmarked against RBI, SEBI, and IT Act standards, ensuring the platform's outputs remain aligned with national regulatory expectations as it scales toward live Account Aggregator integration.

6 System Design

6.1 Hybrid Database Architecture

We Win uses a deliberate two-database design rather than a single general-purpose store, because the two dominant data shapes in the system – high-frequency events and relationship-heavy identity data – are best served by different engines:

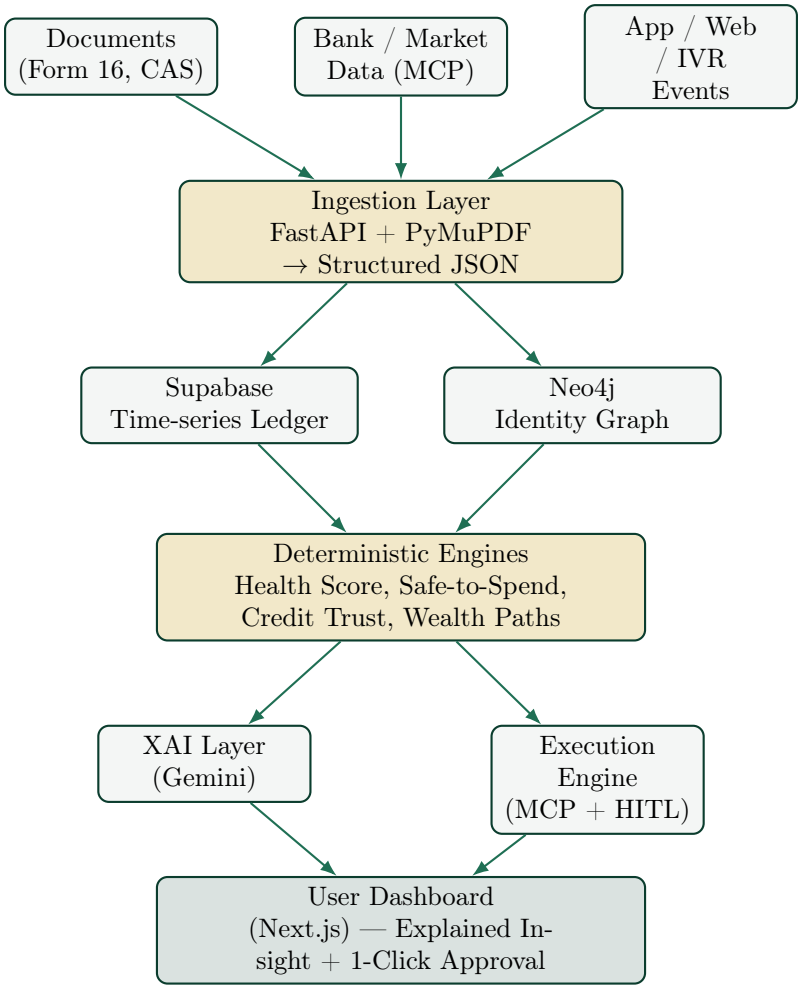
- **Supabase (PostgreSQL)** acts as the primary, high-velocity ledger. It instantly records every mobile app tap, web portal click, IVR call, API timeout, and raw transaction log as an immutable, time-series event.
- **Neo4j** sits directly on top of Supabase as an Identity & Context Graph. It takes the fragmented, high-frequency logs and maps them into a relational knowledge graph – unifying a user's financial profile, family goals, life events, and cross-channel interactions into a single, traversable node.

This split lets We Win answer two very different classes of question efficiently: "what happened, in order, and when" (Supabase) and "how is everything about this user connected" (Neo4j) – the latter being what powers instant incident re-weighting, cross-channel journey stitching, and context-aware market queries.

6.2 Layered Component Architecture

Layer	Stack	Responsibility
Frontend	Next.js	Interactive real-time dashboard; translates AI/engine outputs into plain-English, actionable UI.
Backend Core	FastAPI (Python), PyMuPDF	Document extraction, structured JSON pipelines, RBI/SEBI rule enforcement.
AI & Explainability	Google Gemini, LangChain, CrewAI	Autonomous agent routing to backend calculation tools; explains *why*, never computes *what*.
Data Persistence	Supabase (PostgreSQL)	Immutable, time-series event ledger and JSON state store.
Identity Graph	Neo4j	Unified relational context: profile, goals, incidents, cross-channel journeys.
Market Integration	yfinance, MFAPI, MCP	Scheduled, holdings-scoped market and mutual fund data retrieval.
Execution & Aggregation	MCP	Multi-bank data aggregation, drafted action execution, HITL-gated order placement.

6.3 End-to-End Data Flow



6.4 Design Principles

1. **Separation of computation and cognition:** deterministic Python engines own every number; the LLM layer owns only language – a hard architectural boundary, not a prompting convention.
2. **Event-driven recalculation:** the system never re-runs a full plan on a timer; it reacts to specific events (a new transaction, a life incident, a market move) and recalculates only the affected part of the roadmap.
3. **Human-in-the-loop by default:** every MCP execution tool terminates in a draft, not an action, until the user explicitly approves it.
4. **Zero marginal cost per user:** because the heavy financial logic is algorithmic rather than human-in-the-loop on our side, the platform scales horizontally on cloud infrastructure without adding advisors, support staff, or manual review per additional user.
5. **Compliance-by-construction:** every scoring dimension is benchmarked directly against RBI, SEBI, and IT Act standards at the engine level, so compliance is a property of the calculation itself rather than a downstream audit step.